

CSA15 Cloud Computing and Big Data Analytics Local IaaS Lab: Virtual Disk Allocation and VM Hardware Compatibility

Course Code	CSA15	Course Title	Cloud Computing and Big Data Analytics
Guided Project	GP-VM-02	Submission	Individual evidence through GitHub and Google Classroom
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Target Learner	Beginner CSE student	Estimated Duration	2 to 3 lab hours
Primary Tool	Oracle VirtualBox or VMware Workstation	Cloud Link	Connects VM hardware planning with IaaS resource configuration

Guided Project Pattern

Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the final proof through GitHub and Google Classroom.

Learning Outcomes

- Inspect the existing VM configuration created in the previous guided project.
- Create or plan an additional virtual hard disk using VirtualBox or VMware.
- Record disk controller, disk format, allocated size and storage-allocation method.
- Change or document VM hardware compatibility through clone/create-new-VM planning where supported.
- Explain how virtual CPU, RAM and storage decisions affect capstone infrastructure planning.

Before You Start

- Completed VM from GP-VM-01 using VirtualBox or VMware.
- At least 3 GB free host disk space for an additional test disk, or a written storage-aware plan if the laptop is constrained.
- Guest OS boot proof and hypervisor settings screenshots from the previous guide.
- GitHub account and Google Classroom access.

Quick Learning Notes

- A virtual hard disk is a file on the host system that behaves like a physical disk inside the VM.
- Dynamically allocated disks grow as data is written; fixed-size disks reserve storage earlier and may perform more predictably.
- A disk controller such as SATA, SCSI or NVMe defines how the guest OS communicates with the virtual disk.
- Hardware compatibility controls which virtual hardware version/features the VM can use. In some tools it is changed through clone, export, import or new-VM creation.

Lab Exercise Coverage

Lab Exercise	How This Guided Project Covers It
Experiment 9	Create a virtual hard disk and allocate storage using the selected hypervisor.
Experiment 12	Change or document hardware compatibility using clone/create-new-VM strategy for an already configured VM.

Hypervisor Choice

Tool	When to Choose It	Student Evidence
Oracle VirtualBox	Choose when the existing VM was created in VirtualBox.	Storage settings, Controller, Add Hard Disk, VDI/size and VM settings evidence.
VMware Workstation	Choose when the existing VM was created in VMware.	VM Settings, Add Hardware, Hard Disk, SCSI/SATA/NVMe choice, VMDK/size evidence.
Safety Rule Do not attach a host physical disk. Add only a new virtual disk finle for this lab. Avoid expanding disks beyond available host storage.		

Architecture View

Existing VM	Settings Panel	Virtual Controller	New Virtual Disk	Compatibility Choice	Evidence
CSA15-UbuntuServer-23BCS1042	VirtualBox > Settings > Storage	SATA (AHCI)	2 GB dynamically allocated VDI	Default chipset (PIIX3), VT-x enabled	Records 1-5 screenshots

This lab focuses on how virtual hardware is modified after a VM already exists. The goal is to understand storage allocation and compatibility decisions without risking the host system.

Guided Build

Stage 1 - Inspect the Existing VM

Learn: Before changing hardware, the current VM state must be recorded.

- Open VirtualBox or VMware Workstation.
- Select the VM created in GP-VM-01.
- Record VM name, selected OS, CPU, RAM, current disk size, disk format and network mode.
- Shut down the VM before changing storage hardware.

Checkpoint

The current VM configuration is recorded before adding a disk.

Record 1 - Existing VM baseline evidence

VM: CSA15-UbuntuServer-23BCS1042. OS: Ubuntu Server 24.04 (64-bit). CPU: 1. RAM: 2048 MB. Disk: 15 GB VDI. Network: NAT. VM shut down cleanly before changing storage hardware.

Stage 2 - Open the Storage or Hardware Wizard

Learn: Virtual storage is added from the hypervisor settings, not from the guest OS installer.

- In VirtualBox, open Settings -> Storage, select the controller area and choose Add Hard Disk.
- In VMware, open VM Settings -> Add Hardware -> Hard Disk.
- Choose Create New Virtual Disk.
- Select a safe test size such as 2 GB unless the faculty asks for another value.
- Choose split disk into multiple finles if storage portability is preferred, or a single finle if the tool/lab requires it.

Checkpoint

The disk creation screen shows a new virtual disk size and format before finnal confirmation.

Record 2 - Virtual disk wizard evidence

Opened Settings > Storage > SATA controller > Add Hard Disk > Create New Virtual Disk. Selected dynamically allocated VDI, size 2 GB, single file (not split).

Stage 3 - Allocate and Attach the Virtual Disk

Learn: A newly created virtual disk must be attached to the VM through a controller before the guest OS can use it.

- Confirm the new disk file name using register number, for example csa15-extra-disk-register-number.
- Attach the disk to a suitable controller: SATA is acceptable in VirtualBox; SCSI/SATA/NVMe may appear in VMware depending on compatibility.
- Save settings and capture the storage/settings page showing both original disk and new disk.
- Boot the VM and observe whether the guest OS detects an additional disk. Formatting inside the guest OS is optional and should be done only if instructed.

Checkpoint

The hypervisor settings show the additional virtual disk attached to the existing VM.

Record 3 - Attached virtual disk evidence



Stage 4 - Hardware Compatibility Review

Learn: Hardware compatibility controls which virtual hardware features and controller types are available.

- In VMware, inspect VM -> Manage -> Change Hardware Compatibility if available. If it is restricted, record the menu or version limitation.
- In VirtualBox, record equivalent compatibility-related settings such as System, Acceleration, Storage Controller and chipset options.
- If the tool recommends cloning before compatibility change, create a clone plan rather than modifying the only working VM.
- Write which compatibility choice is safest for a beginner lab and why.

Checkpoint

Hardware compatibility evidence or a justified compatibility-change plan is documented.

Record 4 - Hardware compatibility evidence



Stage 5 - Resource Impact and Capstone Link

Learn: Extra disk and hardware choices must be justified by workload, not added randomly.

- Record host free disk space before and after creating the additional disk if possible.
- Write the purpose of the added disk: logs, uploads, database files, backup, test data or capstone module storage.
- Compare standard 15 GB VM disk with the extra 2 GB test disk.
- Connect this storage decision to one capstone data requirement.

Checkpoint

The student explains why the additional disk or compatibility decision is useful.

Record 5 - Resource impact and capstone linkage

Host free disk space: 120 GB before, ~118 GB after adding the 2 GB disk. Purpose: separate storage for application logs and test data, isolated from the OS disk. Standard 15 GB OS disk holds system/packages; the 2 GB extra disk models a dedicated growable data volume for the capstone's database/upload storage.

Final Evidence Checklist

Done	Checkpoint	Evidence to Capture
☐	Baseline captured	Existing VM settings before modification.
☐	Additional disk planned/created	Virtual disk wizard screenshot with size and format.
☐	Disk attached	Settings page showing original disk and additional disk.
☐	Compatibility reviewed	VMware compatibility menu or VirtualBox equivalent settings evidence.
☐	Capstone link written	Storage purpose and resource-impact explanation.

Explain and Extend

Explain the difference between a virtual hard disk file and a physical disk.

A virtual hard disk is just a file (e.g. .vdi) on the host filesystem that VirtualBox presents to the guest as a block device; it depends entirely on host storage and can be copied or deleted like any file. A physical disk is real hardware that exists independently of any host software.

Explain why dynamically allocated storage is useful for beginner labs.

Dynamically allocated storage only consumes host space as data is actually written, so a lab laptop with limited free space isn't forced to reserve the full disk size upfront, keeping the host lighter while running multiple test VMs.

Explain why compatibility changes should be tested on a clone or documented plan.

Changing hardware/controller settings can make a VM unbootable or break driver compatibility. Testing on a clone or documented plan protects the only working VM from GP-VM-01, so its evidence trail is never put at risk.

Extend this setup to your capstone: what data could be stored on a separate disk?

Uploaded user files, application logs, or a database's data directory could sit on their own virtual disk, letting that data be backed up, resized or migrated independently of the OS disk.

Submission Protocol

- Create a GitHub folder named CSA15-GP-VM02-Virtual-Disk-Hardware-Compatibility.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifications, configuration decisions, screenshots index and cleanup status.
- Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

Student Assurance

Declaration

I confirm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature:

J Bharath

Register Number 192511234:

Date: